An Unsupervised Anomaly Detection Approach for Spacecraft Based on Normal Behavior Clustering

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***Abstract*—This paper presents a new unsupervised anomaly detection approach for spacecraft based on normal behavior clustering. This method takes as input a set of unlabelled historical telemetry data and automatically detects anomalies within the data. After these abnormal data are removed, the method constructs system normal behavior model based on normal data clustering. Then at run-time, it monitors the status of the spacecraft and detects any anomalies appearing in the real-time telemetry data by checking deviations from the normal behavior model. The experimental results show that the method is efficient and practical for anomaly detection of spacecraft system.**

Keywords-Anomaly Detection; Clustering; Unsupervised Learning; Nearest Neighbor Algorithm; Spacecraft

I. INTRODUCTION

With the development of the space industry, modern spacecraft are becoming more advanced and complex. Because of the complex structure of the spacecraft system and the harshness of the space environment, it is practically impossible to completely eliminate the possibility of anomalies or faults, even if we increase the reliability of the system components to the limit. In addition, the space is so distant from the earth that it is extremely difficult to directly repair or replace a damaged component. Therefore, early detection of anomalies or faults in the spacecraft system is significantly important.

Many approaches have been applied to anomaly detection for spacecraft system. Limit checking is the most fundamental and widely used technique. Its basic function is to check whether various telemetry parameters of the spacecraft system are within pre-defined upper and lower limits. Though the limit checking has an advantage that it is simple enough for human operators to implement a system, it lacks flexibility and can not check some small anomalies that occur without violating the limits on the variables.

To overcome the limitations of simple limit checking techniques, a number of intelligent monitoring system such as rule-based expert systems [1] or model-based reasoning methods [2, 3] have been developed. However, these approaches are heavily dependent on a priori knowledge on the system behavior for the spacecraft and have difficulties in acquiring accurate and complete models and knowledge of the spacecraft systems beforehand. Furthermore, these approaches may not be able to recognize anomalies or faults that involve relationships among large numbers of parameters.

On the orther hand, the spacecraft transmits multi-dimensional telemetry data to ground stations everyday. These telemetry data can contain a wealth of information about complex system behavior of the spacecraft. With the recent development of data mining and machine learning technologies, it is possible to examine this archived data and extract embedded information for the anomaly detection and diagnosis [4, 5]. These approaches can also been called the "data-driven" approaches [6, 7]. They seek to build a model for the anomaly detection and diagnosis directly from vast archived spacecraft telemetry data, rather than building it based on human expertise. A significant advantage of these approaches compared with the expert systems and model-based approach is that it does not require complete and accurate expert knowledge or models.

In this paper, we propose a new unsupervised anomaly detection approach for spacecraft based on normal behavior clustering. This method automatically detects and removes abnormal data from the input dataset and builds system normal behavior model by extracting general clusters from the rest normal data. After that, it monitors the status of the spacecraft and detects anomalies appearing in the real-time telemetry data by checking whether these real-time telemetry data deviate from the normal behavior model or not.

The rest of this paper is organized as follows. In section 2, we describe our approach in details. In section 3, we discuss some results of an experiment. Finally, Section 4 concludes the paper and discusses some areas for future work.

II. PROPOSED METHOD

In this section, we describe the proposed framework of our anomaly detection approach, and illustrate how to use large unlabelled historical data to build normal behavior model of the spacecraft system and how to detect anomalies in real-time telemetry data using the model.

A. Overview of the Framework

The framework of our anomaly detection approach is shown in Figure 1. In our framework, the input data is a large amount of archived spacecraft telemetry data which may contain a large number of normal behavior data as well as some of abnormal behavior data. The input data is unlabelled because in practice, labelled data or purely

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normal data is not readily available since it is time-consuming and expensive to manually classify it.

Before processing the data, we first select a list of related parameters for components or subsystems of the spacecraft system which need to been monitored. The values of these parameters form an N-dimensional vector X (x_1, x_2, …, x_n). These vectors can be treated as points in an N-dimensional vector space. Then, we need to detect and remove anomalies from the input data before clustering them. After the anomalies are detected and removed, the normal data are clustered and the normal behavior model is constructed. Finally, at run-time, we attempt to detect deviations from the normal behavior model in real-time observed data.

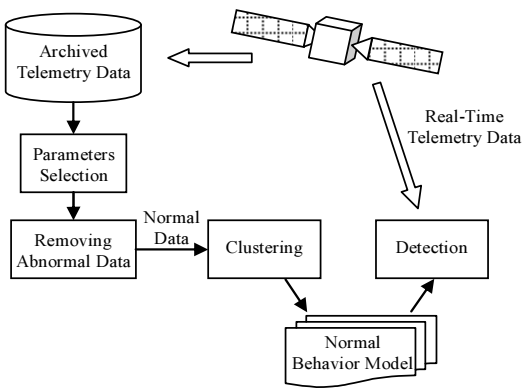

Figure 1. The Framework of Our Approach

B. Parameters Selection

The spacecraft system contains thousands of telemetry parameters. For each component or subsystem of the system which need to been monitored, we first select a list of parameters related to it. These parameters can also contain derived parameters computed from other collected data. The values of these parameters which are collected simultaneously form an N-dimensional vector X (x_1, x_2, …, x_n). These vectors can be treated as points in an N-dimensional vector space. We can define similarity between two vectors in N-dimensional vector space based on appropriate distance metric. The familiar Euclidean distance metric has proven effective in several applications, though other metrics may also be useful.

For two vectors $X_i(x_{i1}, x_{i2}, \ldots, x_{in})$ and $X_j(x_{j1}, x_{j2}, \ldots, x_{jn})$, the weighted Euclidean distance of the two vectors can be calculated as below:

$$dis(X_i, X_j) = \sqrt{w_k \cdot \sum_{k=1}^{n} (x_{ik} - x_{jk})^2} \tag{1}$$

Here, w_k is the weight given to certain vector parameters (Commonly, w_k=1).

Then, the similarity of the two vectors can be defined by the reciprocal of the distance:

$$sim(X_i, X_j) = 1 / dis(X_i, X_j) \tag{2}$$

In archived telemetry dataset, different parameter may have different measure and range. For example, in a given dataset a pressure parameter may range from 0 to 8000 kPa, while a temperature parameter may only cover 10 to 50 degrees. This would imply that a temperature change of one degree is probably more significant than a one kPa pressure change, since the temperature change covers a larger portion of the expected parameter range than the pressure change. Thus, data normalization within the vector is required to avoid disproportionately weighting parameters with differing ranges when calculating distances between vectors.

Z-score transformation is a classical method to normalize data. The Z-score transformation calculates the mean and standard deviation for each parameter from the archived dataset. The normalized vector values are calculated by subtracting the parameter mean from the current value then dividing the difference by the standard deviation.

$$x_i' = \frac{x_i - mean(x_i)}{standard(x_i)} \tag{3}$$

Here *mean*(*) is the mean function and *standard*(*) is the standard deviation of the parameter.

In some cases, it may be advantageous to increase or decrease the weight given to certain vector parameters. For instance, increasing weight of a parameter to have a larger possible *range* relative to other parameters in the vector will amplify any deviations in that parameter. Conversely, decreasing weight of a parameter will reduce the chance of unnecessary alarms when that parameter value changes by an insignificant amount.

C. Removing Abnormal Data

In our method, we use the normal behavior data of the system to build normal behavior model. However, the archived telemetry data is unlabelled. These data may contain some of abnormal behavior data because in practical, it is very hard to guarantee that there are no anomalies during the time we are collecting the data. In addition, it is time-consuming and expensive to manually classify the massive archived data.

To address these problems, we use an unsupervised anomaly detection algorithm called k-Nearest Neighbors (kNN) algorithm. The algorithm makes two assumptions about the data.

- The first assumption is that the number of normal instances vastly outnumbers the number of anomalies.
- The second assumption is that the anomalies themselves are qualitatively different from the normal instances.

Based on the two assumption, we define an anomaly to be a point whose nearest neighbors in data space are far away from it. To measure the distance between two points, we use the weighted Euclidean distance defined in formula (1). The k-Nearest Neighbors determines whether or not a point lies in a sparse region of the data space by computing the sum of the distances to the k-Nearest Neighbors of the point. We refer to this quantity as the kNN score for a point.

Intuitively, the points in dense regions will have many points near them and will have a small kNN score. If the size

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of k exceeds the frequency of any given anomaly type in the dataset and the points of the anomalies are far from the points of the normal instances, then the kNN score can be used to detect these anomalies.

D. Normal Behavior Clustering

After these anomalies are detected and removed, we can then build normal behavior model by clustering the rest normal data in groups.

To create clusters from the rest data, we use a simple variant of single-linkage clustering. The algorithm starts with an empty set of clusters, and updates it as it proceeds. For each new data instance retrieved from the rest normal dataset, we compute the similarity between the data instance and each of the clusters in the cluster set. For a data vector X (x_1, x_2, …, x_n) and a cluster C_j = (X_{j1}, X_{j2}, …, X_{jm}) , the similarity between them can be defined as follow:

$$sim(X, C_j) = \mathrm{Max}(sim(X, X_{jk})) \quad (k = 1,2,\cdots,m) \qquad (4)$$

Based on the above definition, the cluster with the largest similarity (or shortest distance) is selected, and if that similarity is more than some constant α then the data instance is assigned to that cluster. Otherwise, the data instance is dissimilar from any cluster in the set, and therefore a new cluster is created with the data instance as its center. More formally, the algorithm proceeds as follows:

Step 1. Initialize the set of clusters, S, to the empty set.

Step 2. Obtain a data instance (vector) X from the normal data set. If S is empty, then create a cluster with X as the defining instance, and add it to S. Otherwise, find the cluster C in S, such that for all C_i in S, sim(X,C) > sim(X, C_i).

Step 3. If sim(X, C) > α , then associate X with the cluster C. Otherwise, new cluster must be created for it : S ← S {C_n}, C_n is a cluster with X as its defining instance.

Step 4. Repeat steps 2 and 3, until no instances are left in the rest normal data set.

The similarity constant α is usually defined by the reciprocal of the maximum allowed cluster radius. In our method, we randomly select a subset of the training data and compute the mean EX and standard deviation DX of the distances between points in these selected data. The maximum allowed cluster radius is then selected in the range [EX-0.5DX, EX+0.5DX].

After the clusters are constructed, we then compute the centroid O_i and max radius R_i for each cluster.

E. Detection

Thus far, we have constructed the normal behavior clusters for the spacecraft system. Then, for each new data instance in the real-time telemetry data, we detect anomalies by checking deviate from the normal behavior model. As shown in Figure 2, for a data vector X, we compute the distance d between the vector and each of the clusters:

$$d = dis(X, O_i) - R_i \qquad (5)$$

Then, we find the minimum distance d' . If the minimum distance d' more than a distance constant β, the data instance is considered to be an anomaly.

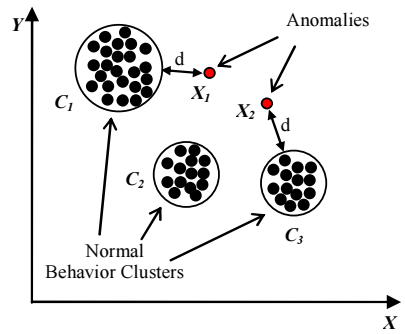

Figure 2. Detecting Anomalies Using Normal Behavior Clusters

III. EXPERIMENTS

We have implemented the anomaly detection approach described above and applied it to the telemetry data of an actual in-orbit satellite. In this section, we describe the experiment dataset and give some experiment results in details.

A. Experiment Dataset

We have applied our approach to the telemetry data of the power subsystem of an actual in-orbit satellite. We picked up 6 parameters relating to the power subsystem of the satellite, which is shown in TABLE I below.

TABLE I. PARAMETERS DESCRIPTIONS OF THE POWER SUBSYSTEM

Parameters	Descriptions
x_1	Battery Charge current
x_2	Battery Discharge current
x_3	Battery Voltage
x_4	Solar Cell Temperature
x_5	Regulator Output Current
x_6	Bus Output Power

The train dataset is some archived historical telemetry data about two weeks. These data is unlabelled and may contain a small amount of abnormal data.

The test dataset is a set of simulated data which contain some simulated fault case.

B. Performance Measures

To evaluate the anomaly detection approach, two major indicators of performance were of interest: the detection rate and the false positive rate. The detection rate is defined as the number of anomalies detected by the system divided by the total number of anomalies present in the test set. The false positive rate is defined as the total number of normal instances that were classified as anomalies divided by the total number of normal instances.

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C. Experiment Results

In our experiment, we first normalized the input data and used the k-Nearest Neighbors algorithm (k=20) to remove abnormal data. Then, we built the normal behavior model by clustering the rest normal data. We selected a subset (about 10%) of the normal data and computed the mean *EX* and standard deviation *DX* of the distances between points in the selected subset. The EX = 0.364, DX = 0.215, so the maximum allowed cluster radius is selected as 0.4 and the similarity constant α is set to 2.5.

After that, we detected anomalies in the test dataset. TABLE II shows the results of the experiments in different distance constant β. Figure 3 shows the ROC (Receiver Operating Characteristic) curve depicting the relationship between false positive and detection rates. From the experiments, we can see that small β have a high detection rate, but the false positive rate is also high. So we must select an appropriate parameter β to compromise between the detection rate and the false positive rate.

TABLE II. THE EXPERIMENT RESULTS OF OUR APPROACH

constant β	detection rate	false positive rate
2	82.8%	1.2%
1.6	83.5%	2.8%
1.2	93.5%	4.6%
1	98.3%	9.7%
0.6	99.2%	14.8%

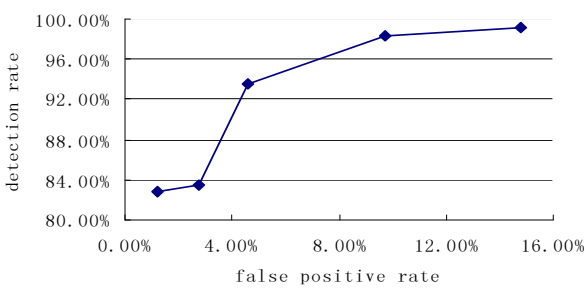

Figure 3. The ROC Curve of False Positive Vs. Detection Rate

One limitation of our approach is that parameter *k* must exceed the frequency of any anomaly type in the archived dataset. If the kNN algorithm could not remove all the anomalies and there are some anomalies contained in the rest normal data, the algorithm may not detect future anomalies of these types because it will assume that they are normal. Thus, in some cases, it needs to remove large anomaly type based on human expertise.

IV. CONCLUSION AND FUTURE WORK

In this paper, we proposed a novel anomaly detection method for spacecraft based on normal behavior clustering. The method uses the k-Nearest Neighbors algorithm to remove abnormal data and build normal behavior model based on the normal data clustering. The result shows that the method can successfully detect anomalies in the real-time telemetry data underlying the normal behavior model.

The contributions of this paper are as follows.

- The proposed method builds a model for anomalies detection directly from archived data, rather than building it based on human expertise. Thereby, it requires little a priori knowledge and can attain a low construction cost.
- The proposed method is unsupervised and can train on unlabelled data. Thereby, it can reduce training data requirement and is capable of detecting previously "unknown" anomalies.
- In contrast to traditional individual parameter monitoring schemes, the proposed method is capable of characterizing and monitoring interactions between multiple parameters and can provide additional insight for the spacecraft system.

There are some directions for future works. In this paper, the anomaly detection method does not make use of any expert knowledge. We plan to explore ways of using background knowledge within the automated anomaly detection context. The semi-supervised learning algorithms can be used as one possible way to make use of this knowledge. In addition, the datasets used in this research are all time series. Our method, however, do not explicitly make use of time. Instead, time is ignored and the method treats each single point in time as a data point. In future work, we plan to find a way to detect anomalous subsequences in multivariate time series.

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